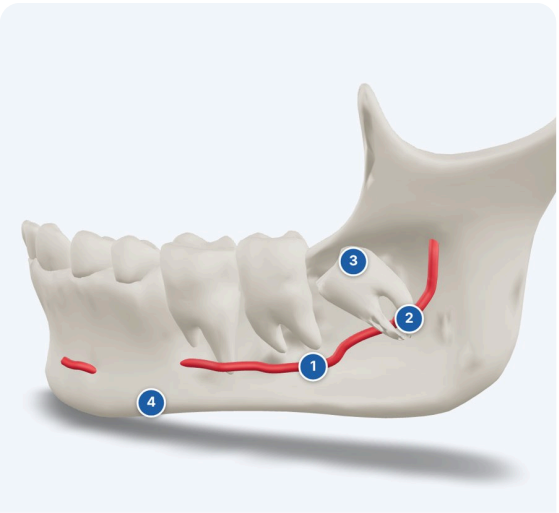
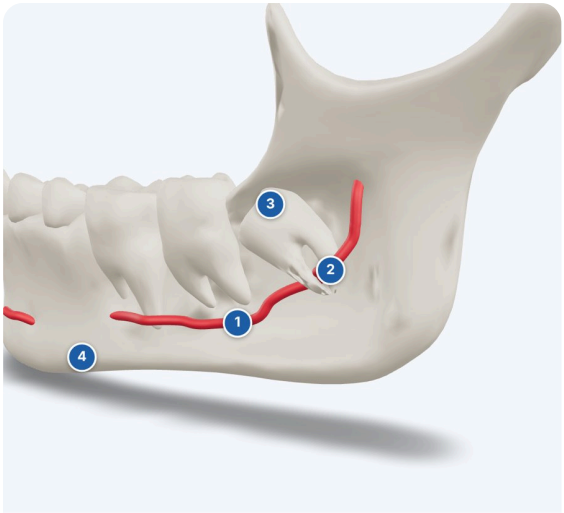
Nagoya University

Impacted lower wisdom tooth and the nerve beside it



Whole jaw



Close-up of the roots and nerve

1. Nerve and artery 2. Roots of the wisdom tooth 3. Wisdom tooth 4. Jaw bone

Jaw bone and teeth Nerve and artery (in their canal)

What is the problem?

An **impacted lower wisdom tooth** is stuck under the gum and bone. It cannot grow in the right way.

The nerve

A nerve runs inside the lower jaw. Its name is the **inferior alveolar nerve**. An artery (a blood vessel) runs next to it. The nerve gives feeling to the lower lip, chin, lower teeth and gums on the same side.[1]

In this model, **the roots of the wisdom tooth wrap around the nerve and artery**.

Why taking the tooth out can hurt the nerve

To take the tooth out, the roots must come out of the bone. The roots curve around the nerve. So the nerve can be pressed, stretched or cut as the roots come out. The artery can also bleed.

What nerve damage feels like

The lower lip, chin, teeth or gums on that side may feel numb or tingly, or just feel different. Sometimes they hurt. The lip and face can still move as normal.[1]

Short-term: the feeling comes back. This most often takes weeks to months.[1]
Long-term: the feeling does not fully come back.[1]

How often does it happen?

Some wisdom teeth have roots close to the nerve. A review of 16 studies looked at these teeth. It found:[2]

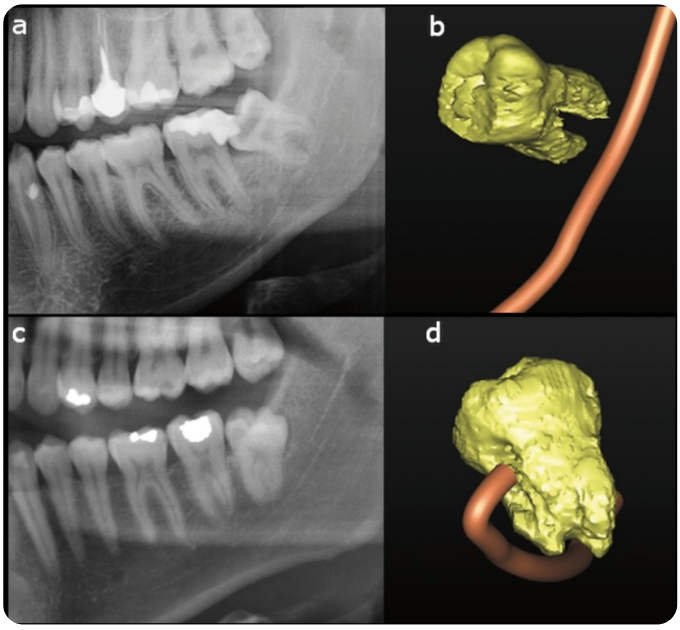
About 7 in 100
had nerve damage after the whole tooth was taken out

About 1 in 100
had nerve damage that did not go away

Why a 3D scan can help

The usual dental X-ray of the whole mouth is called an **OPG**. It is a flat picture. On a flat picture, the roots can look like they touch the nerve, even when they do not.[3] Some signs on an OPG can warn that the roots are close to the nerve.[4]

A **CBCT** scan is a 3D X-ray. It can show if the roots really touch the nerve. It also shows which side of the roots the nerve is on.[3] Studies show it does this better than an OPG.[5]



The two OPGs (a and c) look much the same. The CBCT images show they are not. Top (b): the tooth does not touch the nerve. Bottom (d): the roots wrap around the nerve. In b and d, the tooth is yellow and the nerve is orange.[3]

Picture: Rodriguez y Baena R, et al. J Clin Exp Dent. 2017, Figure 1. Licence: CC BY 2.5. Not changed.

A CBCT scan uses more X-ray than an OPG. So it is used when an OPG shows the roots may be close to the nerve.[3]

Choices to talk about with a surgeon

- **Leave the tooth in and check it often**, if it is not causing problems.
- **Coronectomy:** take out only the top of the tooth. Leave the roots around the nerve. In the same review, about 1 in 200 people had nerve damage after this.[2] But the roots can move. Once in a while they need to come out later.
- **Take out the whole tooth.**

Questions to ask the surgeon

- How close are the roots to the nerve?
- What is the risk of nerve damage in this case?
- Is a CBCT scan needed?
- Is a coronectomy a good choice in this case?
- What happens if the tooth is left in?

Sources

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This leaflet is for a Nagoya University research study and for general information. It is not personal medical advice. A dentist or oral surgeon can give advice on each person's situation.